

Anonymization & Exchange Gateway for Imaging Studies

External Browser Upload · DICOM Network (C-STORE SCP, port 11112) · Programmatic Batch Ingest

- Drag-and-drop: files, folders, DICOM directories
- Client-side tag de-identification (DICOM PS3.15)
- Anonymization preview — before / after tag diff
- Multi-study detection with per-study upload cards
- Per-project retained-tag anonymization profiles
- Auto-retry: 3x exponential backoff per file

- Compute Engine VM — aegis-prod-dimse-receiver
- Static IP 35.232.172.221 - TCP port 11112
- C-STORE SCP — receives from PACS systems / scanners
- Institution attribution (AE title or AE CIDR)
- Calls POST /api/ingest on DICOM association close
- Durable retry queue + dead-letter (disk-persistent)

- Only tag-de-identified DICOM is transmitted
- HIPAA Safe Harbor — 18 identifier types removed
- Deterministic UID hashing + date shifting
- Zero software install required at sending site
- DIMSE path ingests raw files (internal-only path)

aegis-prod-488120 · us-central1 · Live, February 2026

Cloud Armor (DDoS / WAF) + Global HTTPS Load Balancer + Identity-Aware Proxy (admin routes) + Managed SSL cert v3 (api · admin · aegisimaging.ai · www) + Cloud Run (API · Dashboard · Landing · 10 sidecars)

- Upload orchestration (sessions, chunked PUT)
- Study / project / institution management
- Routing rules engine → auto-pipeline dispatcher
- DICOMweb proxy (QIDO-RS + WADO-RS)
- Export shares: token-auth, ZIP download
- Email digest scheduler + SMTP relay
- DIMSE retry control proxy
- MCP tool backend + batch import CLI
- distroless image → minimal CVE surface

- Behind Identity-Aware Proxy (IAP)
- Study browser: filter, search, paginate, bulk ops
- Weasils DWV — yoked before/after defacing review
- 7-stage pipeline visualization per study
- RBAC: admin (write) + viewer (read-only)
- Routing rules, institutions, anon profiles
- Protocol templates, API keys, webhooks
- Audit log + CSV export, share management
- Federation peers, project lifecycle

- Studies, projects, institutions, admin users
- Routing rules, audit trail, export shares
- Private IP · Secret Manager credentials
- Point-in-time recovery (7-day retention)

- `dicom/raw/{uid}/` – tag-de-identified
- `dicom/clean/{uid}/` – defaced + processed
- `bids/{uid}/` – NIFTI / BIDS output
- `analytics/{uid}/` – segmentation + metrics
- Shared volume (Go API + all 10 sidecars)

- Head MRI / PET / CT — facial feature removal
- mri_reface / DeepDefacer / mri_deface backends
- nibabel fallback for dev / testing only
- Writes defaced DICOM to clean/ store
- SSIM-based QA score per study

- Burned-in text OCR on DICOM pixel data
- Gemini / Cloud Vision / Azure / Textract / Tesseract
- Pixel redaction — detect + mask burned-in PHI
- LLM-based text scrubbing (Gemini / OpenAI / Anthropic)
- Per-project confidence threshold config

- File integrity + required DICOM tag check
- Slice consistency (rows / cols / pixel spacing)
- SNR estimation (signal mean / corner noise)
- Coverage completeness by body part
- Missing slice gap detection

- Fills modality + body_part from DICOM headers
- Heuristic: tags → SOP UID → series description
- Gemini / Cloud Vision / Rekognition fallback
- Re-evaluates routing rules after classify
- Confidence threshold gating (≥ 0.5)

- DICOM → NIfTI via dcm2niix
- BIDS-compliant directory structure
- sub-{hash8}/anat | func | dwi | perf | pet
- JSON sidecar metadata per series
- Privacy: UID-hashed subject labels

- Verifies TR / TE / flip / thickness vs template
- Classic + Enhanced DICOM (multi-frame)
- Per-project protocol templates with CRUD
- numeric / exact / range / contains_all match
- critical / warning / info severity levels

- Brain: FreeSurfer · SynthSeg · BrainSuite · volBrain · Atlas ROI
- Spine: TotalSpineSeg · SPINEPS
- Whole-body: TotalSegmentator · nnU-Net · MedSAM2 · MONAI Label
- Specialized: PETSURfer · QSM · BASIL · FSL · ANTs · SPM · ITK-SNAP
- Longitudinal: TBM-SyN + FreeSurfer Long stream

- Spinal cord segmentation (sct_deepseg_sc)
- Vertebral labeling (sct_label_vertebrae)
- CSA per vertebral level (sct_process_segmentation)
- Compression metrics: aMCC, aSCOR
- DTI mapping: FA, MD, AD, RD (optional DWI)

- Synthetic brain MRI generation
- CPU: Shepp-Logan phantom (numpy + pydicom)
- GPU: MONAI BraTS LDM (INCLUDE_MONAI=true)
- T1w contrast, Rician noise, face anatomy
- Seeded + reproducible, 20 – 200 slices

- VPC private subnets + Cloud NAT
- Cloud Armor DDoS / WAF
- TLS 1.2+ on all endpoints
- CMEK (Cloud KMS)
- IAM least privilege
- Secret Manager (DB creds, API keys)
- Cloud Audit Logs
- Artifact Registry vuln scanning
- distroless containers (Go API)
- IAP on all admin routes
- No PHI in email / audit entries

- Token-authenticated share links
- Study info, expiry countdown (server-anchored)
- ZIP download of approved DICOM files

- DICOMweb STOW-RS destinations
- DIMSE C-STORE to remote AE Title
- Auto-forwards on study approval
- Cross-cloud: GCP↔AWS STOW-RS live

- Extend / revoke anytime
- Immutable download log per share
- Export analytics dashboard

- 96 read tools: studies, audit, stats, routing, health
- 95 write tools (confirm:true + reason required)
- Zod-validated schemas — read-safe by default
- DIMSE retry proxy: process, replay, dead-letter

- DICOM tag provenance analysis
- Diagnostic tools for study triage
- Cohort report, pipeline funnel, project health

- aegis-import CLI — bulk historical migration
- 5 webhook events, HMAC-SHA256 signed payloads
- Machine-to-machine bearer tokens (hashed)

- aegisimaging.ai — public marketing site
- React + Vite built, served by nginx
- Interactive synthetic DICOM demo widget
- Schedule Demo / Contact form
- Market opportunity: \$45B imaging market
- LB default backend — no IAP required
- www + apex domains on SSL cert v3

301691475234 · us-east-1 · ✓ Live — aws.api.aegisimaging.ai · Cross-cloud routing verified · Feb 25, 2026

- HTTPS listener on ACM certificate
- Cognito hosted UI — admin-create-only user pool
- authenticate-cognito default action
- Public bypass rules: /healthz, upload, export

- Go API — same image as GCP (1 vCPU / 2 GB)
- Admin Dashboard — React / nginx (0.5 vCPU / 1 GB)
- Service discovery: api.aegis.local
- Force-new-deployment via GitHub Actions

- defacing · phi-detection · qc-service
- bids-service · classification-service
- protocol-service · synth-service
- analytics-service · sct-service
- Cloud Map private DNS: svc.aegis.local:8080

- RDS PostgreSQL 15 — private subnet
- Secrets Manager — master credentials
- S3 DICOM bucket — versioned, KMS-encrypted
- Same STORAGE_MODE=s3 as GCP (same Go code)

- Elastic IP — stable for PACS AE title registration
- Amazon Linux 2023 — Docker + SSM agent
- SSM Parameter Store → image URI on every boot
- GitHub Actions: write SSM param + reboot instance

- Triggers on push to develop — parallel with GCP Cloud Build
- Matrix build: 11 services, --platform linux/amd64
- Push SHA tag + latest tag to 13 ECR repositories
- Force-new-deployment for 10 ECS services
- Update SSM param + reboot DIMSE EC2

Container Apps · PostgreSQL Flexible Server · Azure Blob Storage · ✓ Live — Feb 26, 2026 · GitHub Actions OIDC CI/CD

- ACR — private registry, GitHub Actions OIDC push
- Easy Auth on Container Apps — injects X-MS-CLIENT-PRINCIPAL-NAME
- AUTH_PROVIDER=azure, AUTH_ENABLED=true
- Federated OIDC — no long-lived credentials stored

- Go API — same image as GCP/AWS (0.5–1 vCPU / 2 GB)
- Admin Dashboard — React / nginx Container App
- Azure Communication Services SMTP relay (smtp.azurecomm.net)
- Force-new-revision via GitHub Actions on every push

- defacing · phi-detection · qc-service
- bids-service · classification-service
- protocol-service · synth-service
- analytics-service · sct-service
- Internal ingress only — same Docker images as GCP/AWS

- Azure Database for PostgreSQL — Flexible Server
- Azure Blob Storage — STORAGE_MODE=azure
- DefaultAzureCredential — Workload Identity / Managed Identity

- Debian 12 — Docker + Azure VM Extensions
- Static public IP — TCP port 11112 for PACS registration
- GitHub Actions: az vm run-command + docker pull/restart

- Triggers on push to develop — parallel with GCP + AWS
- Federated OIDC — AZURE_CLIENT_ID / TENANT_ID / SUBSCRIPTION_ID
- Build 13 images → push to ACR → az containerapp update

Automated Processing Pipeline

routing rules define which steps are required → pipeline dispatches each step in dependency order, hands-free

1. Classify

fills modality
body_part

2. PHI Scan

OCR pixels
redact burned-in

3. Protocol

TR / TE / flip
vs template

4. Deface

remove face
head imaging

5. QC Check

SNR · coverage
slice consistency

6. BIDS

NIfTI + sidecar
BIDS structure

7. Analytics

18 backends
brain · spine · body

8. SCT

cord CSA
compression

9. Export Fwd

STOW-RS or
DIMSE forward

Implementation Phases

M1: Foundation + GCP ✓

Go API · 9 Python sidecars · Upload Portal · Admin Dashboard · PostgreSQL · Terraform · CI/CD

M2: AWS Live ✓

ECS Fargate · RDS · S3 · ALB + Cognito · Cross-cloud DICOM routing

M3: Azure Live ✓

Container Apps · PostgreSQL Flex · Azure Blob · ACR · GitHub Actions OIDC CI/CD

M4: Processing & Security ✓

JPEG2000 / Enhanced DICOM · Pixel redaction · Clinical trial RBAC · Security hardening

M5: Analytics & MIDI-B ✓

18 analysis tools · SCT · Longitudinal · Biomarker DB · MIDI-B compliance · Niivue

M6: Private Beta •

Invite-gated access · Feedback · DIMSE C-MOVE / C-FIND · SLA hardening

M7: Compliance ◦

BAA template · SOC 2 Type I audit · DIMSE at scale · AWS Marketplace

M8: Enterprise Integrations ◦

SOC 2 Type II · HL7 FHIR · Federation · DICOM conformance v2

M9: Enterprise GA ◦

On-premises · Multi-tenant SaaS · PACS/VNA native integration

Key Open-Source Dependencies

Go:

suyashkumar/dicom · pgx · testcontainers-go · testify

Browser:

dcmjs · dicomParser · Weasis DWV (MIT) · React 19 · Vite

Defacing:

mri_reface · DeepDefacer · mri_deface · dcm2niix · nibabel

PHI/OCR:

pytesseract · Google Cloud Vision · AWS Textract · Pillow

QC/BIDS:

pydicom · numpy · dcm2niix · pyntdicom (C-STORE SCP)

Analytics:

FreeSurfer · FSL · ANTs · SPM · SynthSeg · nnU-Net · TotalSegmentator · MONAI Label

SCT/Spine:

Spinal Cord Toolbox · TotalSpineSeg · SPINEPS · MedSAM2

Synth:

numpy · pydicom · MONAI Generative · torch · huggingface-hub

Infra:

Terraform Google + AWS + Azure providers · Docker distroless/slim

Multi-Cloud + Test Coverage

GCP:

Cloud Run · Cloud SQL · GCS · IAP · Cloud Armor · KMS · Pub/Sub

AWS:

ECS Fargate · RDS · S3 · ALB + Cognito · ACM · Secrets Manager

Local:

Docker Compose · PostgreSQL 15 · Mailpit · local filesystem

Auth:

GCP IAP · Azure AD Easy Auth · AWS ALB+Cognito · dev auto-auth

Storage:

STORAGE_MODE=gcs | s3 | azure | local — same Go API, no code changes

Azure:

Container Apps · PostgreSQL Flex · Azure Blob · ACR · Easy Auth · Azure VM (DIMSE)

CI/CD:

GitHub Actions: Go (1,072) · Python (892) · Client (166) · TS · Docker (15) · Cloud Build auto-deploy

Domains:

aegisimaging.ai · www · api · admin — SSL cert v3